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4302-Project 02: Complete Product Prototype

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Case Study

Prototype Link

https://www.figma.com/proto/kssIcHKxdIqqcT0Rl3ecFt/4302-Final-Project--Copy-?node-id=2002-80&p=f&t=ewj16pKUETyveMts-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=2002%3A80

Problem and Context

Real-World Problem

After a demanding workday, decision fatigue can create a mental barrier that makes it difficult to transform the available ingredients in your kitchen into nutritious meals. This disconnect often leads to impulsive and pricey panic orders, even when your pantry is fully stocked.

Who experiences it

Individuals in high-cognitive-load professions, such as leaders, executives, and healthcare providers, often experience significant mental energy depletion by evening. This fatigue leaves them ill-equipped to meet domestic demands, such as planning dinner, organizing household chores, or managing family schedules.

Why it matters

If left unaddressed, this issue causes a cycle of financial stress and worsening health. It makes the kitchen a source of frustration and leads to the waste of ingredients. Decision fatigue can weaken self-control and cause impulsive choices.

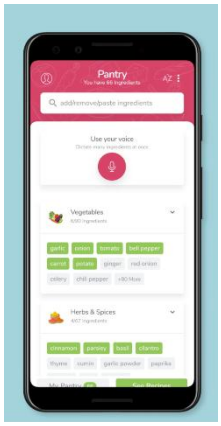
Personal connection

I find myself caught in the tension between my desire for a healthier lifestyle and the reality of mental fatigue. I truly value efficiency and resourcefulness; however, I'm watching my pantry go to waste while my bank account and energy levels take a hit due to my reliance on takeout that could easily be avoided.

Research and Insights

Analysis of Existing Solutions

- **SuperCook (Recipe-by-Ingredient Engine)**



Note. From SuperCook - Recipe Generator (Version 2.0.29.1) [Mobile app], by SuperCook, 2025, Google Play Store (play.google.com). Screenshot by author.

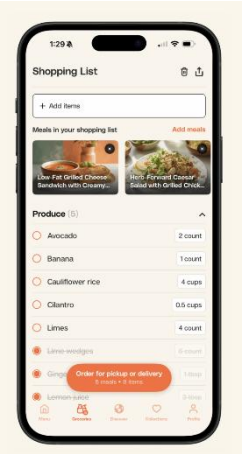
  - **Description:** Introducing a search engine and app that allows users to easily input or voice their available ingredients. In return, they can discover thousands of recipes they can whip up right away with what they have on hand (Larson, 2025).
  - **Strengths:** This tool tackles the common excuse of "there's nothing to eat" by uncovering hidden options. It features filters for meal type, cooking time, and specific ingredients to use, such as wilting produce.
  - **Weaknesses:** It takes a considerable amount of time and effort, up to 30 minutes, to input all the items from your pantry, and it's crucial for users to keep it updated while they cook.
  - **Target User:** Home cooks on a budget, who have well-stocked pantries but feel too mentally drained to come up with new combinations of staples, often need a little inspiration.

- **Samsung Food (AI-Powered Smart Kitchen Platform)**

Note. The image illustrates the synchronization between the Family Hub refrigerator screen and the "Food List" on a mobile device. From *Bespoke Family Hub refrigerator and Samsung Food app* [Photograph], by Samsung, 2023, Samsung Newsroom (https://news.samsung.com/global/samsung-announces-global-launch-of-samsung-food-an-ai-powered-personalized-food-and-recipe-service). Copyright 2023 by Samsung.

  - **Description:** Previously known as "Whisk," this platform leverages Vision AI to identify ingredients from photos and connects with smart appliances to streamline cooking processes, such as automatically preheating ovens (Samsung Electronics, 2023).
  - **Strengths:** Lowers the barrier for manual input by utilizing photo-based ingredient recognition and simplifies the process from "concept" to "cooking" by communicating with smart kitchen devices.
  - **Weaknesses:** Full benefits require a modern "smart kitchen" ecosystem (e.g., Samsung appliances); the "Food+" subscription is required for advanced features.
  - **Target User:** Professionals who are well-versed in technology have already embraced smart home ecosystems.

- **Ollie / PlanEat AI (Done-for-You Automated Planners)**



Note. The interface shows a meal-based shopping list with automated categorization of produce like avocado, banana, and cilantro. From *Ollie AI: Family Meal Planner* (Version 1.0.102) [Mobile app], by Confabulation Corporation, 2026, App Store (ollie.ai). Screenshot by author.

  - **Description:** AI-driven apps that craft a realistic weekly meal plan based on your cooking time, family preferences, and budget, all while providing a one-tap grocery list for convenience (Ollie AI, 2025).
  - **Strengths:** It proactively addresses decision fatigue by offering a ready-made plan to follow instead of beginning with a blank search bar. Additionally, it prioritizes ingredient reuse throughout the week to minimize waste.
  - **Weaknesses:** The focus is mainly on making new grocery purchases instead of syncing in real-time with the current pantry inventory.
  - **Target User:** Busy professionals and families seeking a clear, structured plan can benefit from a solution that eliminates the hassle of sifting through countless recipes.

- **KitchenPal / PantryAI (Inventory-First Management)**

Note. The interface displays a digital "Fridge" inventory, sorted by expiration date, showing items such as sour cream, milk, and tomatoes. From *KITCHENPAL: Pantry Inventory* (Version 6.0.4) [Mobile app], by KitchenPal, 2026, Google Play Store (https://play.google.com/store/apps/details?id=fr.icuisto.icuisto). Screenshot by author.

  - **Description:** Applications that focus mainly on inventory tracking can analyze your consumption habits and recommend recipes based on items that are nearing their expiration dates (KitchenPal, n.d.).
  - **Strengths:** Effectively addresses the utility disconnect by monitoring shelf life and raising awareness of available resources before they spoil.
  - **Weaknesses:** Keeping your pantry in check can turn into just another task on your to-do list; true automation is tough to achieve since it depends on users to log the items they've used or to scan every single receipt.

  Target User: Homes that are well-organized prioritize zero-waste objectives and emphasize high financial efficiency.

Synthesis & Market Gaps

Common Patterns:

- **Automation Shift:** Many contemporary solutions are shifting away from basic recipe databases and embracing suggestion engines that require little effort from users.
- **Grocery Integration:** Most leading solutions now connect directly with retailers like Walmart and Amazon Fresh, effectively removing the need for physical shopping.

Remaining Gaps:

- **Real-Time "Passive" Inventory:** As of now, there is no comprehensive solution that effortlessly tracks pantry levels without requiring manual scans or logs. Once a meal is prepared, the system often becomes less accurate unless the user takes the initiative to update it.
- **Contextual Fatigue Awareness:** While many apps focus on nutrition and budgeting, very few take into consideration the fluctuating energy levels of users. For instance, an app could automatically recommend a quick 10-minute meal on a day when the user has a full calendar packed with meetings.
- **The Repurposing Gap:** Many apps concentrate on creating a recipe from the ground up; however, few effectively recognize and recommend ways to utilize specific leftovers or partially used ingredients without requiring manual input.

Existing solutions

Several types of solutions for meal planning and inventory management already exist. *SuperCook* is a recipe-by-ingredient engine that helps users find recipes based on items they already have. Its main weakness is the manual effort required to input and update ingredients. *Samsung Food* is an AI-powered smart kitchen platform that uses Vision AI to identify ingredients from photos and connect with smart appliances, but the full benefits require a dedicated "smart kitchen" ecosystem. *Ollie/PlanEat AI* are automated planners that craft weekly meal plans and grocery lists, proactively addressing decision fatigue, but often prioritize new purchases over existing inventory.

KitchenPal/Pantry AI focuses primarily on inventory tracking and recipe recommendations for items nearing expiration; however, manually updating inventory can become just another task.

User perspectives

Research highlighted two key user perspectives. The high-intensity professional experiences "ego depletion" by evening, finding the simple task of browsing the pantry overwhelming. They are less worried about the cost of takeout and more annoyed by the illogic, frustrated that they can handle complex work problems but not a simple chicken-and-rice dinner. The housewife/parent of three deals with cognitive overload and the emotional strain of anticipating a child's rejection of meals. For this user, the burden of rotting groceries weighs heavily on their conscience, and takeout becomes a means of maintaining household peace rather than simply satisfying hunger.

Opportunity

The primary opportunity lies in bridging the gap between having a stocked kitchen and the mental paralysis caused by post-work decision fatigue. The market is missing a passive, context-aware engine that transforms the interaction from asking "What do you want?" to suggesting, "Given your energy level and what's in your fridge, how about this?" The blending of work and home life is making decision fatigue a pressing concern, and consumers are actively seeking solutions that automate daily choices. The initial approach will prioritize a user-friendly interface that accounts for available groceries and provides a single, high-probability meal suggestion tailored to the user's current fatigue level.

Key insights

The research synthesis revealed several key insights:

- **Automation Shift**: Many contemporary solutions are moving away from basic recipe databases toward suggestion engines that require little user effort.
- **Real-Time "Passive" Inventory:** A comprehensive solution is missing that effortlessly tracks pantry levels without requiring manual scans or logs.
- **Contextual Fatigue Awareness:** Few apps consider the fluctuating energy levels of users to recommend quick meals on demanding days.

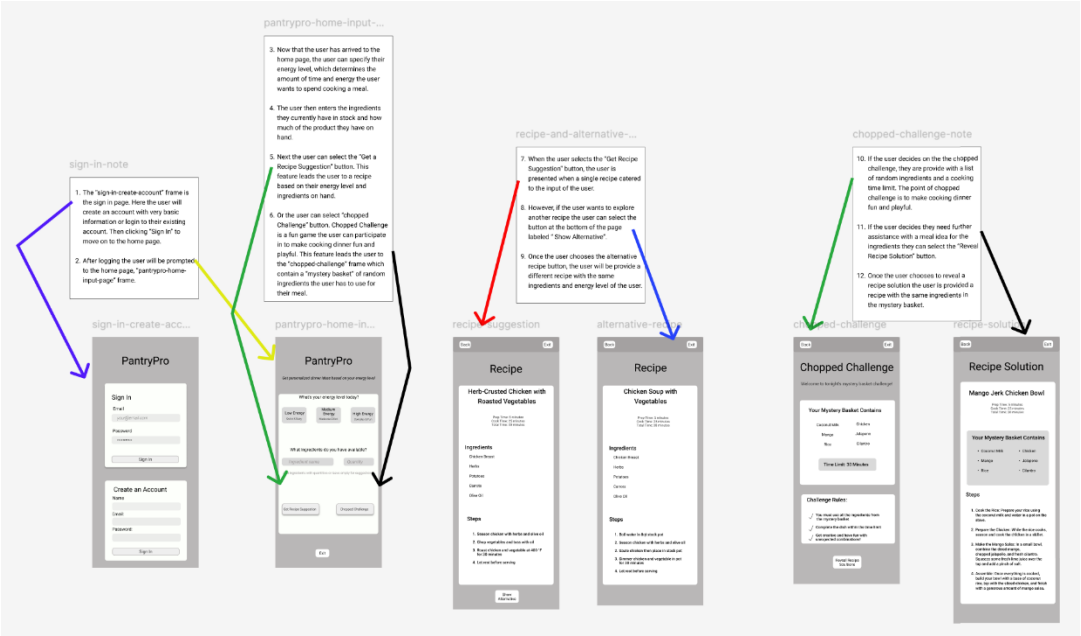
- **The Repurposing Gap**: Few apps effectively recognize and recommend ways to utilize specific leftovers or partially used ingredients without manual input.

Concept Description

Concept overview

The proposed solution is a simple-to-use application designed to help users at any energy level make healthier food choices based on the ingredients they already have in their kitchen. By analyzing the contents of users' pantries and fridges, the app provides tailored meal suggestions that promote nutritious eating while minimizing food waste. This process streamlines meal planning from hours to a 10- to 30-minute experience, empowering users to embrace culinary creativity and explore new cuisines using ingredients they already own.

The user experience is intuitive, starting with an easy login process. Users manually enter items in stock, filter by availability, and specify current energy levels to receive personalized suggestions. A playful "mystery basket" interface recontextualizes the task as a game, and if an initial suggestion is unsuitable, an alternative recipe is offered.



Core value proposition

The main benefit is saving valuable time and money by eliminating the need for last-minute grocery trips and encouraging the use of existing stock. By utilizing existing groceries, food waste is minimized, and decision fatigue is mitigated, enabling users and their households to make healthier, less stressful dinner choices.

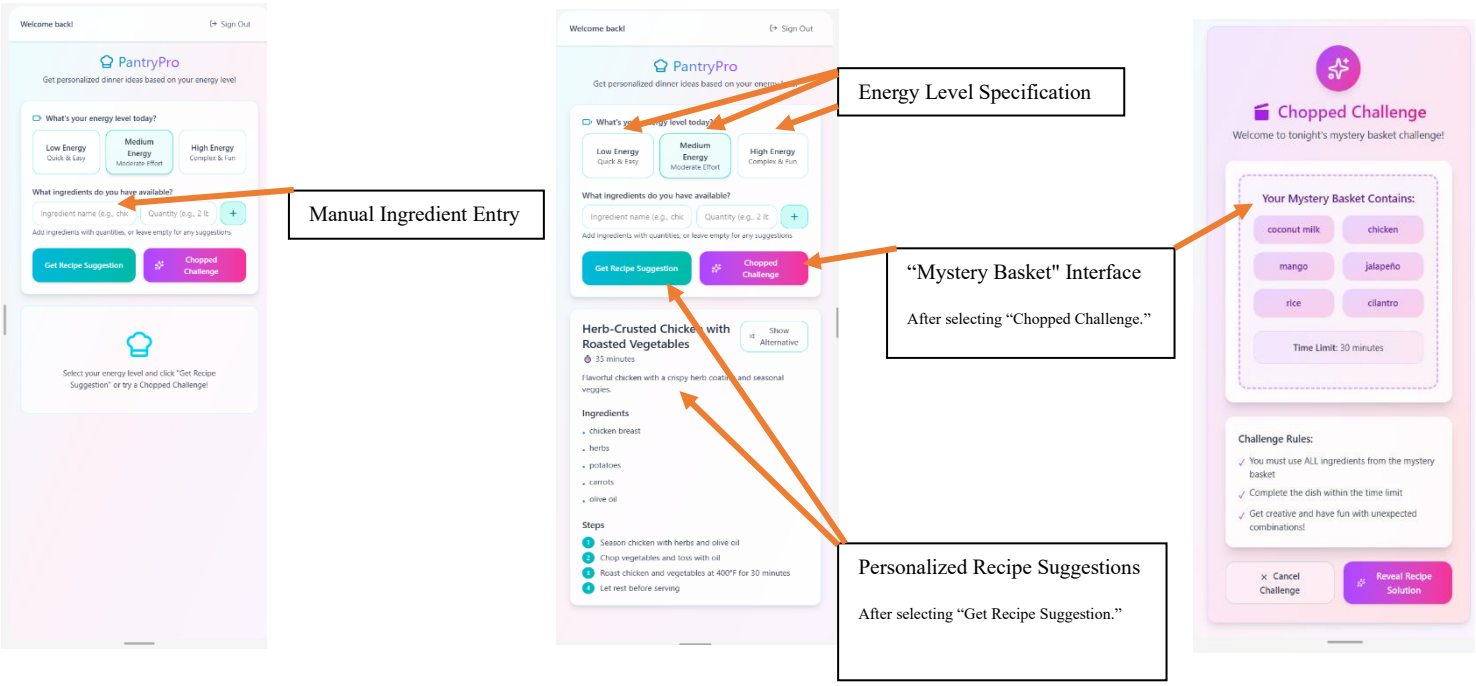
Design Decisions and Rationale

•**Manual Ingredient Entry:** Users can input specific food items currently in stock. This helps the user focus only on the items they plan to use for dinner, rather than cataloging the entire pantry, reducing the number of unwanted tasks.

• **Energy Level Specification:** The app considers the user's current energy level as a crucial factor for tailoring recipe suggestions.

• **Personalized Recipe Suggestions:** The system generates a single, tailored dinner recommendation that accounts for both available ingredients and the selected energy level.

• **"Mystery Basket" Interface:** A playful game-like feature that presents available ingredients as a fun challenge.



User Feedback and Iteration

Part 1: Feedback Analysis

Participant 1: Healthcare Administrator

Note. From Burnout, stress and medical with nurse on floor of hospital for sad, mental health and headache [Photograph], by PeopleImages, n.d. (peopleimages.com). Copyright by PeopleImages.

Input Friction

- **What happened:** The participant let out a sigh when they arrived at the manual entry screen. They began typing “chicken” and “rice” but paused, stating, "Even entering 'broccoli' feels like a hassle right now. Can't I just snap a photo instead?" This response highlighted that manual typing is a major obstacle and leads to considerable frustration.
- **Why it happened:** This represented a missed opportunity to enhance user experience and highlighted an issue with input friction. After a long day, the user anticipated a more streamlined input method, such as image capture, to align with their low energy levels. The reliance on manual entry not only conflicted with the app's primary goal of minimizing decision fatigue, but also added unnecessary complexity to the process.

Single-Choice Success

- **What happened:** The user visibly relaxed upon pressing the "Low Energy" button and took 30 seconds to read through the recipe suggestion for "Herb-Crusted Chicken," nodding in approval afterward.
- **Why it happened:** This illustrates the success of providing a single choice by alleviating the overwhelming paralysis that can occur when dealing with too many options. By offering a carefully chosen suggestion, the decision-making process became more straightforward for an "ego-depleted" user, allowing them to concentrate better.

Participant 2: The Parent

Note. This image depicts the concept of "technoference," where digital devices interrupt parental interactions. From *Mom with a smartphone digital addiction children want to spend time with mom but can not* [Photograph], by N. Galitskaya, 2022, Freepik (https://www.freepik.com/premium-photo/mom-with-smartphone-digital-addiction-children-want-spend-time-with-mom-can_28137227.htm). Copyright 2022 by Natalia Galitskaya.

Feature Misinterpretation

- **What happened:** The participant eagerly tapped the "Chopped Challenge" Button, mistakenly believing it would lead to a quick meal. However, this feature was typically regarded as a main navigation tool instead of a lighthearted "playful twist."
- **Why it happened:** This situation arose from a misunderstanding of the mental model. The user saw the feature's label and position as a shortcut for a primary action rather than as a secondary, optional function.

Single-Choice Success

- **What happened:** The user conveyed their satisfaction with receiving just one suggestion, remarking, "If I had to sift through a list of 50 recipes, I'd simply close the app and order pizza instead."
- **Why it happened:** Just like Participant 1, this highlights how single-choice success can effectively manage cognitive overload. By offering limited options, it provided clear direction and helped the user avoid feeling overwhelmed by too many choices.

Confusing Labels

- **What happened:** The participant was unsure whether "Energy Level" indicated their own personal energy level or was related to the complexity or cooking time of the dish.
- **Why it happened:** The label "Energy Level" was confusing, resulting in misunderstandings about its intended function. This ambiguity created a hierarchy issue that contributed to the misinterpretation.

Part 2: Prioritized Triage List

Category A: Critical Usability Issues

These challenges have a direct effect on the main user experience and have been recognized as major obstacles to the app's success.

- **Manual Input Friction:** The significant effort needed for manual typing often leads to frustration for exhausted users, hindering their ability to complete the essential task of entering ingredients.
    - **Planned Change:** I plan to introduce a "Quick-Add Tags" feature for frequently in-stock items, such as eggs, rice, and pasta. This will allow users to effortlessly add ingredients with just a single tap, eliminating the need for typing.
- **Ambiguous "Energy Level" Labeling:** Users are unclear whether "Energy" pertains to their own fatigue or to the difficulty of the recipe.
    - **Planned Change:** I will rename the energy levels to time-based labels "Quick (15 min)," "Standard (30 min)," and "Creative (45+ min)" to clearly communicate cooking duration.
- **Feature Misinterpretation (Chopped Challenge):** Users often confuse a secondary "playful" feature for a primary navigation shortcut.
    - **Planned Change:** I will reorganize the visual hierarchy to highlight the main flow as the key path, while clearly indicating that the "Chopped Challenge" is an optional activity.

Category B: Preference-Based Feedback

These items are improvements to the user experience that do not prevent users from completing tasks.

- **Request for Image Capture:** Participant 1 proposed the idea of snapping a photo of the ingredients. While this would be a great "nice-to-have" feature, it's important to first enhance the manual entry process by implementing quick-add tags.

• **Gamification (Visual Confirmation):** Incorporating animations or “success” states can transform the experience of completing an entry into more of a game than a mundane task. This approach focuses on addressing emotional responses rather than just tackling functional obstacles.

• **Single-Choice Success Sentiment:** Both users appreciated the single-recipe suggestion; However, this serves more as a validation of a design concept instead of an issue that needs addressing.

Part 3: What I Will Not Change

Based on the feedback analysis, I have decided not to act on the following suggestion:

• **The Suggestion:** Camera-Based Ingredient Entry (Image Capture) Participant 1 proposed, "Why can't I just snap a picture?" to completely skip the typing hassle.

• **Why I will NOT act on it:** Although the idea of photo-to-ingredient technology is certainly appealing, I’ve decided not to move forward with its implementation at this time for two main reasons:

o **Complexity and Scope**: Developing a dependable computer vision system to recognize different food items is a substantial technical challenge that extends beyond the current focus of refining the main process.

o **More Efficient Immediate Solution:** The "Quick-Add Tags" for common ingredients effectively tackle the core issue of input friction identified in the synthesis.

This approach offers a simple yet impactful solution that cuts down on manual typing while avoiding the potential errors and high costs associated with a camera-based system. By emphasizing the use of tags, I am prioritizing a thoughtful design fix that immediately lowers the barrier for users who may feel overwhelmed or low on energy.

Solutions in Operation

Change #1- Manual Input Friction Reduction

- **What feedback prompted it:** Participant 1 expressed frustration with manual typing for ingredients, sighing at the screen and asking for an image capture option instead, highlighting input friction after a long day.
- **What you are changing (or plan to change):** I plan to introduce a "Quick-Add Tags" feature for frequently in-stock items like eggs, rice, and pasta, allowing users to add ingredients with a single tap, eliminating typing.
- **Why this approach:** This approach was chosen as a more efficient, immediate solution than the alternative of a camera-based system. Developing a reliable computer vision system is a substantial technical challenge beyond the current scope. The tags effectively tackle the core issue of input friction, prioritizing a simple fix that immediately lowers the barrier for users who may be overwhelmed or low on energy.

Change #2- Clarifying "Energy Level" Labeling

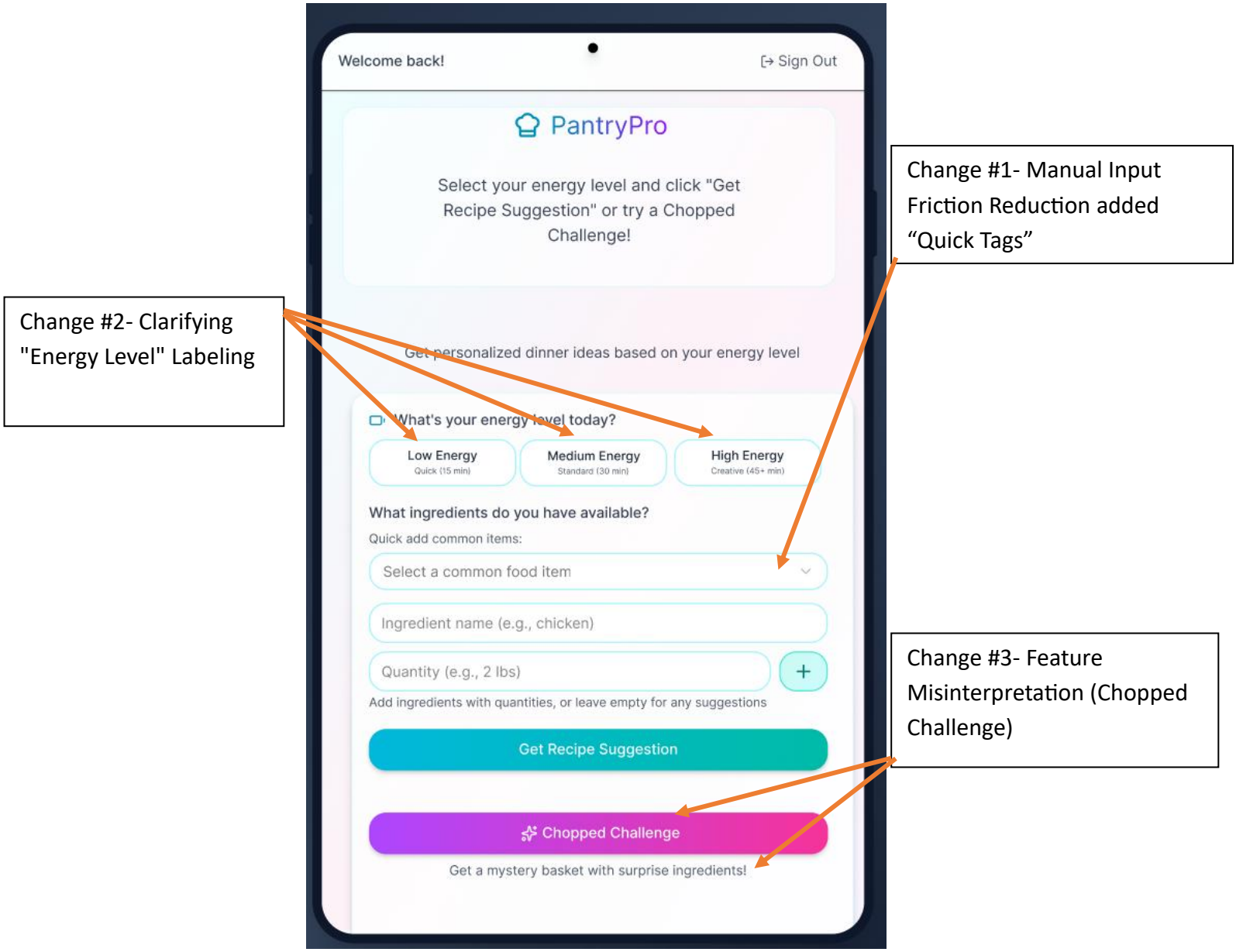
- **What feedback prompted it:** A participant was unsure whether "Energy Level" referred to their own personal energy or the cooking complexity/time of the dish.
- **What you are changing (or plan to change):** I will add explanation above the energy levels to time-based labels: "Quick (15 min)," "Standard (30 min)," and "Creative (45+ min)".
- **Why this approach:** Renaming makes the button's function clear, directly communicating the cooking duration. The fix was obvious because the uncertainty stemmed directly from the label itself, and time is a universally understood measure for recipe selection.

Change #3- Feature Misinterpretation (Chopped Challenge)

- **What feedback prompted it:** A participant eagerly tapped the "Chopped Challenge" button, mistakenly believing it was a quick meal shortcut, when it was intended as a secondary, "playful twist" feature.

- **What you are changing (or plan to change):** I will reorganize the visual hierarchy to highlight the main flow as the primary path, clearly indicating that the "Chopped Challenge" is an optional activity.

- **Why this approach:** The situation arose from a misunderstanding of the user's mental model, where label and position suggested a primary action shortcut. Reorganizing the visual hierarchy is a direct solution to visually separate primary and secondary actions, aligning the interface with user expectations and preventing further misinterpretation.



Reflection

About your problem — What do you understand now that you didn't at the start?

At the begining of this project, I didn't fully grasp how deeply decision fatigue could affect someone responsible for meal preparation. I soon realized that this fatigue serves as a significant mental hurdle, making it difficult for individuals to tackle the intricacies of planning and cooking meals. Moreover, my investigation into available apps uncovered a surprising wealth of options for recipe suggestions and pantry management. However, I noticed a glaring gap: there are very few tools that take a user's personal situation into account, such as their available time, energy levels, and specific dietary needs. Most apps simply offer recipe ideas without considering the real-life constraints and preferences of the user. This insight has led me to a greater appreciation for the importance of a genuinely user-centered approach in creating effective meal planning solutions.

About design — What did this project teach you about design?

This project underscored that design goes beyond just visual elements; it involves crafting a logical framework to enhance user experience. Effective design merges functionality with usability, ensuring that users' needs and interactions take center stage throughout the entire process. I also realized the significance of articulating ideas clearly to AI tools like Figma Make. These tools are incredibly helpful for overseeing the complete process of designing an app, from its initial concept all the way to its final stages.

About yourself — How have you grown as a designer?

The project has experienced significant growth, largely thanks to the valuable insights gathered from real users. Although receiving constructive criticism during the development of personal ideas can be tough, such feedback is crucial for professional advancement and success. This input has greatly enriched our understanding of how design influences user experience, often taking precedence over the technical aspects that developers tend to focus on. By actively engaging with user feedback, we can foster ongoing improvement, ultimately leading to more successful outcomes in any project.

Looking forward — If you continued this project, what would you do next?

To boost user engagement and enjoyment during meal preparation, it would be advantageous to further develop the chopped challenge feature by adding a few more functionalities. A valuable improvement could be the inclusion of a timer within the app. This feature would help users stay focused on the challenge without having to leave the app, thus creating a smoother gaming experience and increasing overall participation. Additionally, integrating a grocery list that links directly to the user's preferred grocery store would be another helpful enhancement. This functionality would simplify the meal preparation and shopping process, reinforcing the idea of an all-in-one app to provide users with a more efficient and enjoyable experience.

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